

LMITO THERAPEUTICS Inc.

Novel Therapy for Neuromuscular Diseases

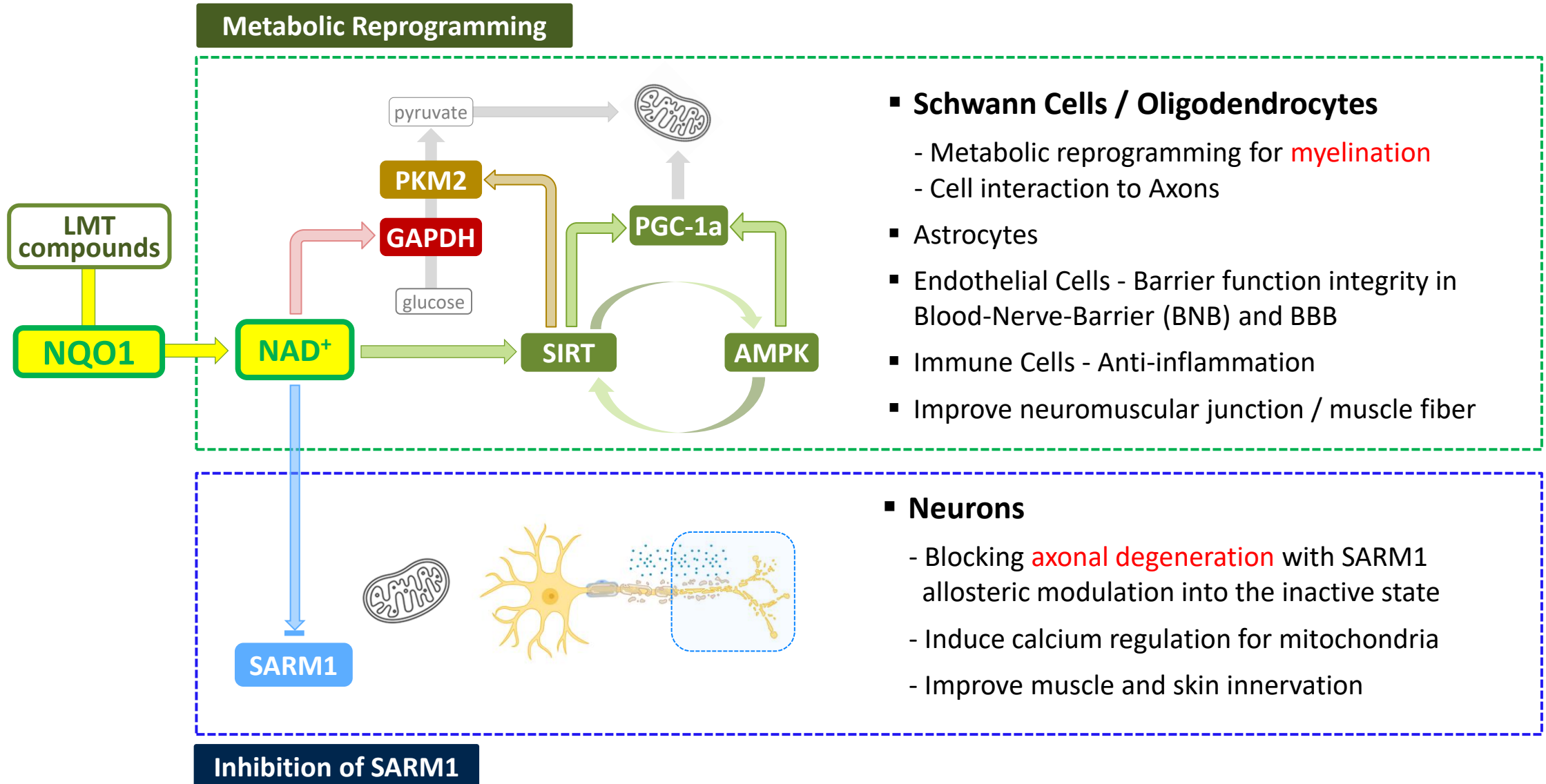
LMT837 / SARM1 / ALS

**Wheeseong Lee, Ph.D.
CEO / Founder**

Representative Developmental Pipeline at Lmito Therapeutics

Target (Mechanism)	Asset	Indication	Stage	Key Highlights
Oxido-reductase	LMT503	Autoimmune Disease (IBD)	Phase 1	• First-in-class metabolic reprogramming drug • Anti-inflammation/-fibrosis, barrier function restitution • CTA (IND) for Phase 1 approved by EMA
	LMT801	Neuropathy (CMT1A)	IND-enabling (GLP-tox)	• Prevents both demyelination and axonal degeneration • Orphan Drug Designation granted by FDA for CMT1A • GLP-tox study completed (KDDF-grant funded)
	LMT837	Neuromuscular Disease (ALS)	IND-enabling (pre-tox)	• Improves motor units via mitochondrial enhancement and SARM1 inhibition • Ready to initiate GLP-tox study • Concurrently mitigates fibrosis and inflammation
PKM2	LMT3012	Metabolic Disease	IND-enabling	• Targets PKM2 “Moonlight function” • IND-enabling study ongoing • Composition-of-matter patent filed

Lmito's Approach for Neurological Disorders

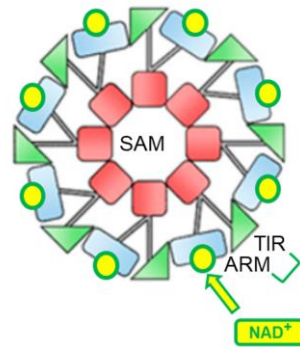


Lmito's Therapy for SARM1 Inhibition - Inducing Inactive Conformation of SARM1

Normal Condition

- SARM1 stays inactive under normal conditions.
- NAD^+ binds to the ARM domain, keeping the TIR domain locked and inactive.

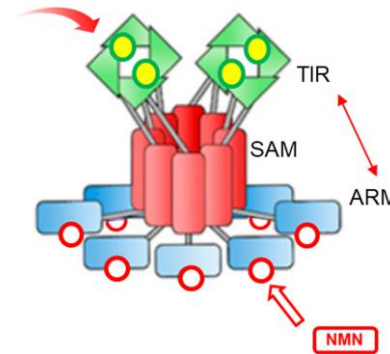
SARM1 inactive state
top view



$\text{NAD}^+ \ll \text{NMN}$

Pathological activation

SARM1 active state
side view



Pathological Condition

- When NMN/ NAD^+ ratio increases, NMN replaces NAD^+ on the ARM domain.
- This activates TIR, depletes NAD^+ , and causes axonal and neuronal degeneration.

$\text{NMN} \ll \text{NAD}^+$

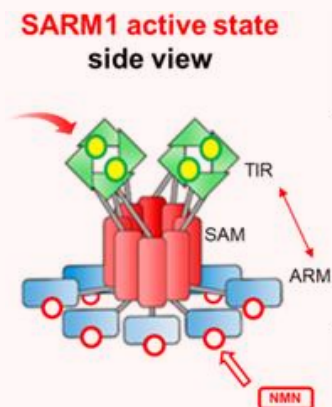
Substrate inhibition

Lmito's Therapeutic Concept

- Our approach increases NAD^+ , which stabilizes SARM1 in its inactive form.
- This prevents TIR activation and blocks neuro-degeneration.

BEI vs Lmito's Allosteric Inhibition Approach

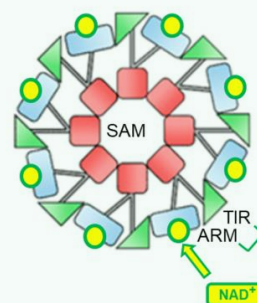
Inhibiting Active Domain Approach with Base Exchange Inhibitors



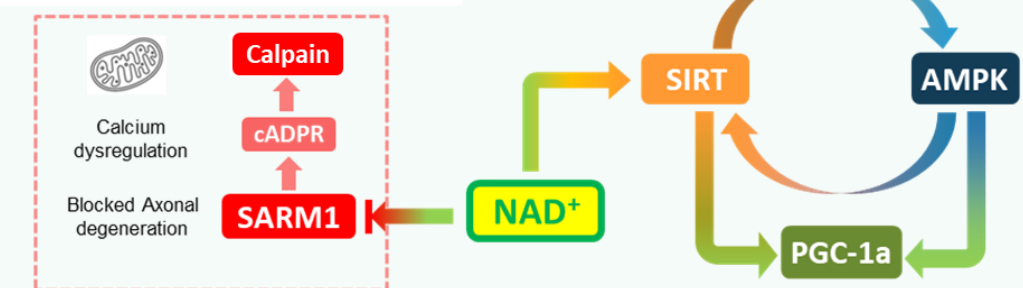
- Risk of paradoxical activation
- Inhibition of physiological function of SARM1
- blocking NAD^+ consumption only
- may not fully restore axon integrity

Lmito's Approach of Substrate Inhibition with NAD^+ through Maintaining SARM1 as an Inactive State

SARM1 inactive state top view

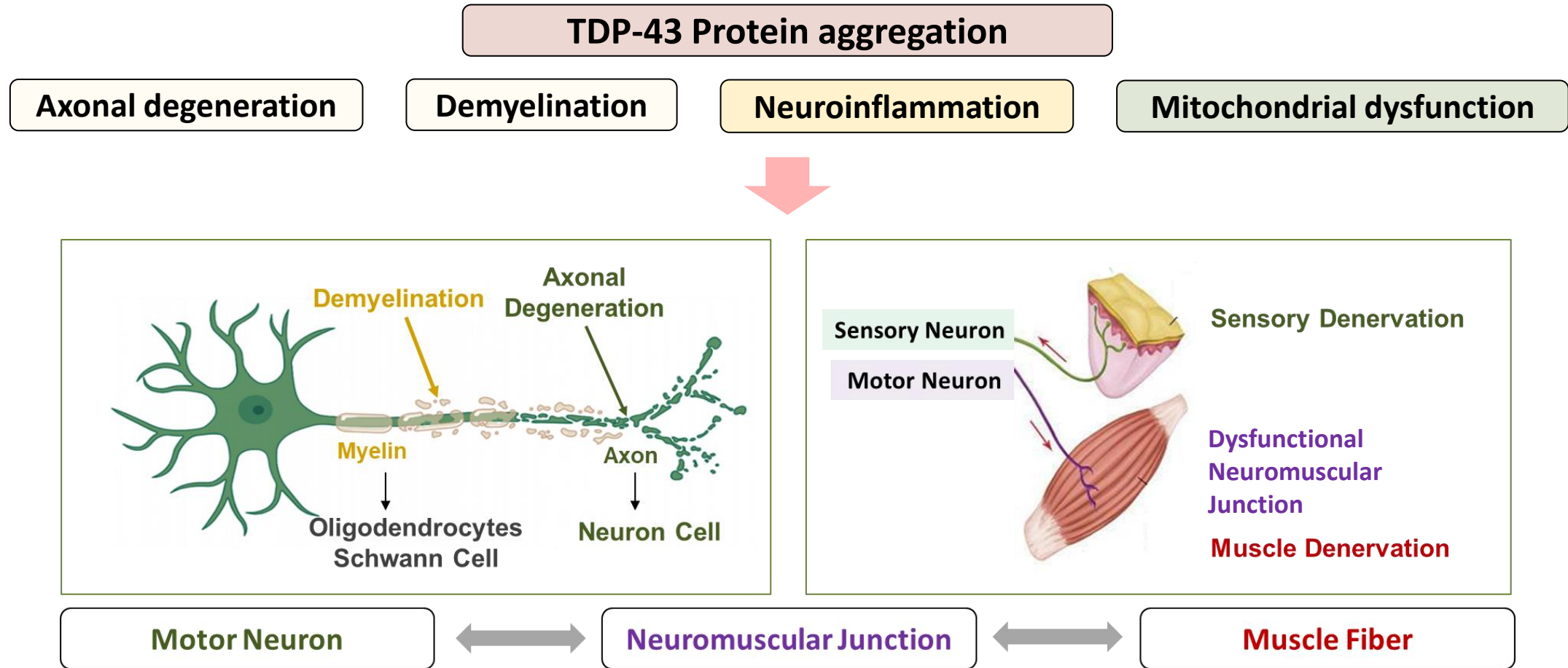


Allosteric SARM1 Modulation



- **Safe substrate inhibition strategy** avoiding total suppression of SARM1 but maintaining as an inactive state
- **Enhanced NAD^+ levels provide dual benefits** : not only blocking NAD^+ consumption but also improving mitochondrial biogenesis, enhanced cellular metabolism, and superior neuroprotection through enhanced level of NAD^+

Lmito's Strategy for Developing ALS Therapy



Reprograms **metabolically over-activated cells within the neuro-microenvironment** to restore mitochondrial integrity, while **allosteric inhibition of SARM1** prevents axonal degeneration - resulting in robust therapeutic efficacy in ALS.

Amyotrophic Lateral Sclerosis and SARM1

Enrichment of *SARM1* alleles encoding variants with constitutively hyperactive NADase in patients with ALS and other motor nerve disorders

Jonathan Gilley^{1*}, Oscar Jackson¹, Menelaos Pipis², Mehrdad A Estiar^{3,4}, Ammar Al-Chalabi^{5,6}, Matt C Danzi⁷, Kristel R van Eijk⁸, Stephen A Goutman⁹, Matthew B Harms¹⁰, Henry Houlden², Alfredo Iacoangeli^{5,11,12}, Julia Kaye¹³, Leandro Lima^{13,14}, Queen Square Genomics², John Ravits¹⁵, Guy A Rouleau^{3,4,16}, Rebecca Schüle¹⁷, Jishu Xu¹⁷, Stephan Züchner⁷, Johnathan Cooper-Knock¹⁸, Ziv Gan-Or^{3,4,16}, Mary M Reilly², Michael P Coleman^{1*}

Axons Matter: The Promise of Treating Neurodegenerative Disorders by Targeting SARM1-Mediated Axonal Degeneration

Raul Krauss,^{1,*} Todd Bosanac,¹ Rajesh Devraj,¹ Thomas Engber,¹ and Robert O. Hughes¹

Abnormal brain protein abundance and mRNA expression of SARM1 in amyotrophic lateral sclerosis

Axon Biology in ALS: Mechanisms of Axon Degeneration and Prospects for Therapy

Michael P. Coleman¹ 

Constitutively active SARM1 variants that induce neuropathy are enriched in ALS patients

A. Joseph Bloom^{1*} , Xianrong Mao¹, Amy Strickland¹, Yo Sasaki¹, Jeffrey Milbrandt^{1*} and Aaron DiAntonio^{2*}

Sarm1 deletion suppresses TDP-43-linked motor neuron degeneration and cortical spine loss

Matthew A. White^{1†}, Ziqiang Lin^{1,2†}, Eugene Kim³, Christopher M. Henstridge⁴, Emiliano Pena Altamira¹, Camille K. Hunt¹, Ella Burchill¹, Isobel Callaghan¹, Andrea Loreto⁵, Heledd Brown-Wright⁶, Richard Mead⁶, Camilla Simmons³, Diana Cash³, Michael P. Coleman^{5,7} and Jemeen Sreedharan^{1*} 

The SARM1 axon degeneration pathway: control of the NAD⁺ metabolome regulates axon survival in health and disease

Matthew D Figley¹ and Aaron DiAntonio^{1,2}

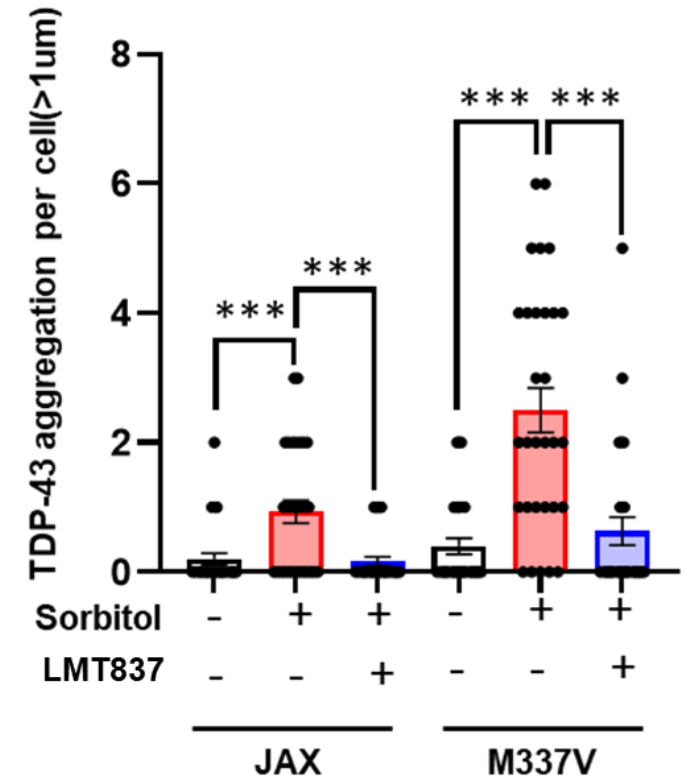
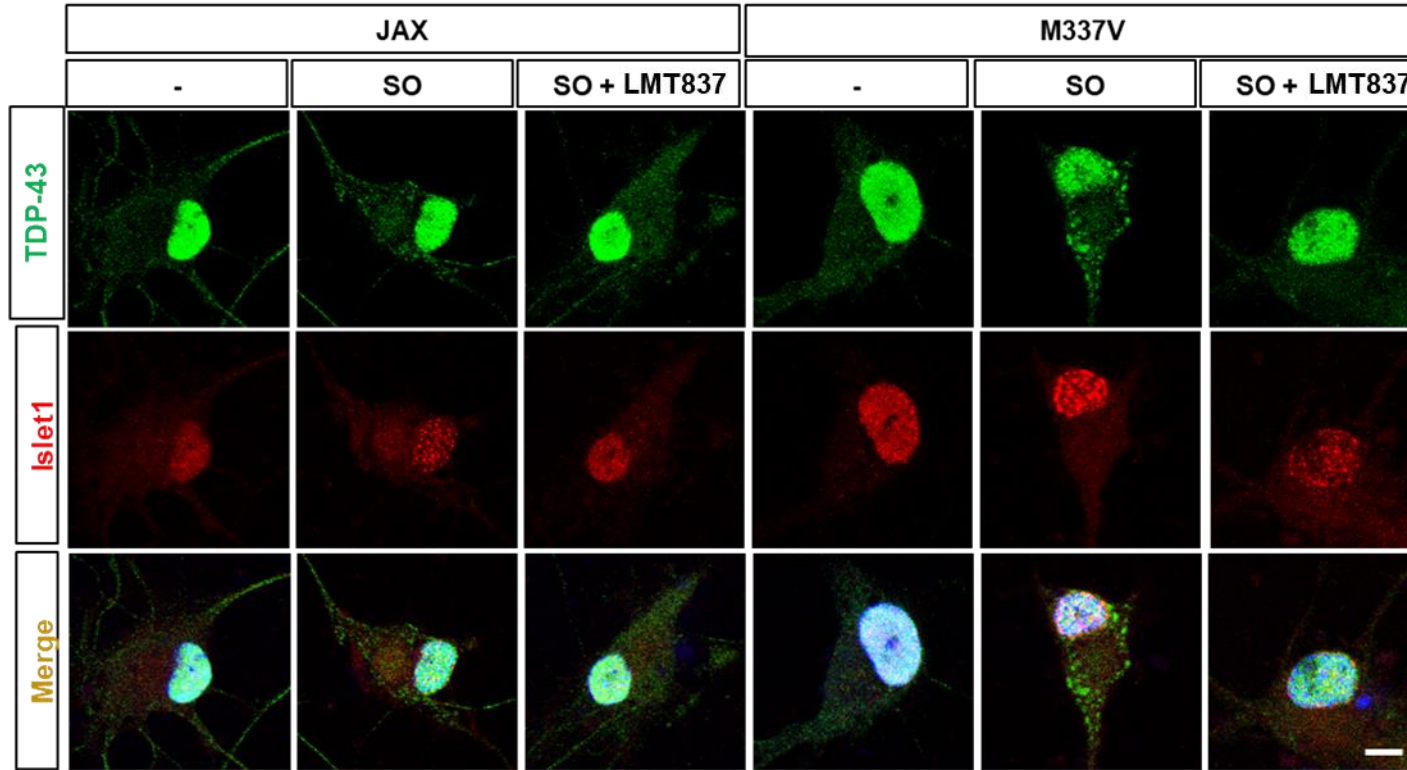
LMT837, Summary of Efficacy

in vitro study

Human iPSC Motor neurons Motor Neuron-like Spinal Cord Cells Hela Cells Microglia and Macrophages Human Epithelial Cells Human Patient-derived Fibroblasts	• Reduced Stress Granule Formation (G3BP1, TIA1) • Reduced TDP-43 Cytosolic Aggregation • Reduced Oxidative Stress • Improved Mitochondrial Function • Mitigated Inflammation and Fibrosis (Cytokines, EMT, ECM) • Induced Metabolic Reprogramming
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LMT837, in vitro Study using Human iPSC

Human induced pluripotent stem cells



Osmotic stress with sorbitol in **iPSC-derived motor neuron lines** : led to cytoplasmic TDP-43 aggregation. LMT837 demonstrated remarkable efficacy in modulating TDP-43 dynamics. NAD⁺ modulation can effectively modulate TDP-43 pathophysiology in **human motor neurons**.

LMT837, Summary of Efficacy

in vivo study

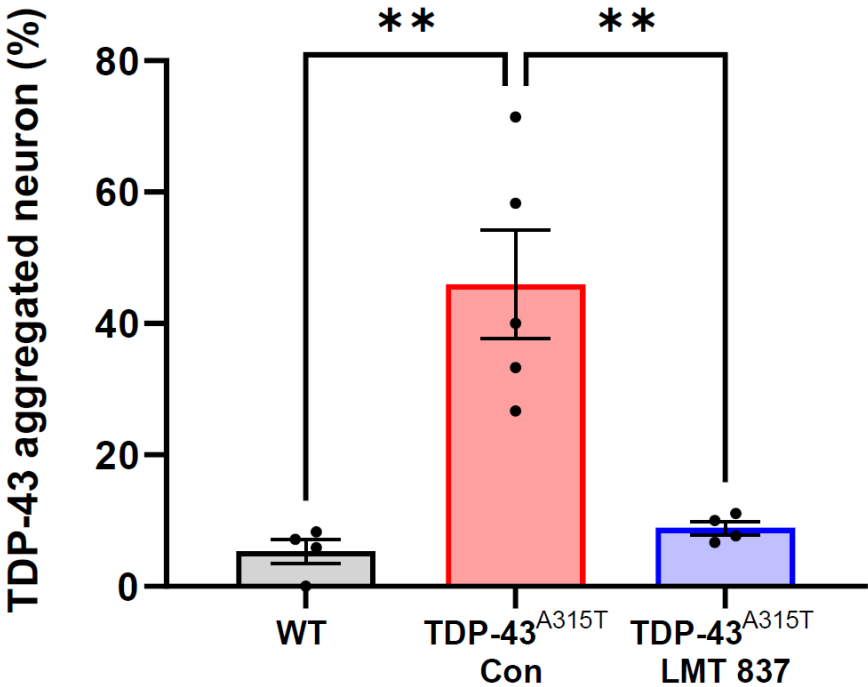
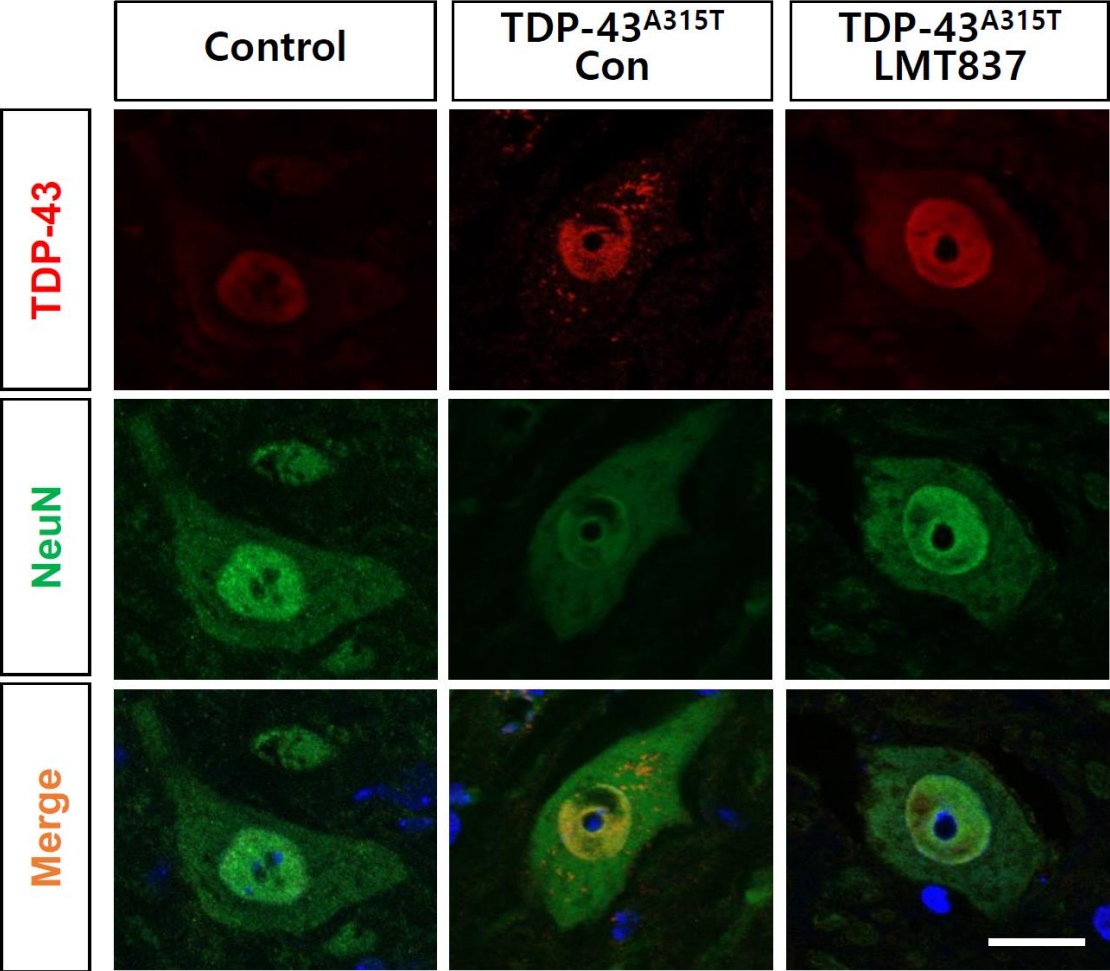
TDP-43*^{A315T} mice [B6SJL-Tg(Prnp-TARDBP*^{A315T})23JleI/J]

Started dosing of LMT837 at day 50 for up to day 180

- Delayed Disease Onset
- Extended Survival
- Improved Motor Function (Rotarod Performance, Grip Strength)
- Increased Motor Neuron Counts but Decreased Dysmorphic Neurons in the Motor Cortex and the Spinal Cord
- Reduced Microglia and Astrocytes Activations (Iba1, GFAP)
- Reduced Cytosolic TDP-43 Aggregates in the Spinal Cord

LMT837 Reduces TDP-43 Aggregation in Spinal Motor Neurons in TDP-43* A315T ALS Model

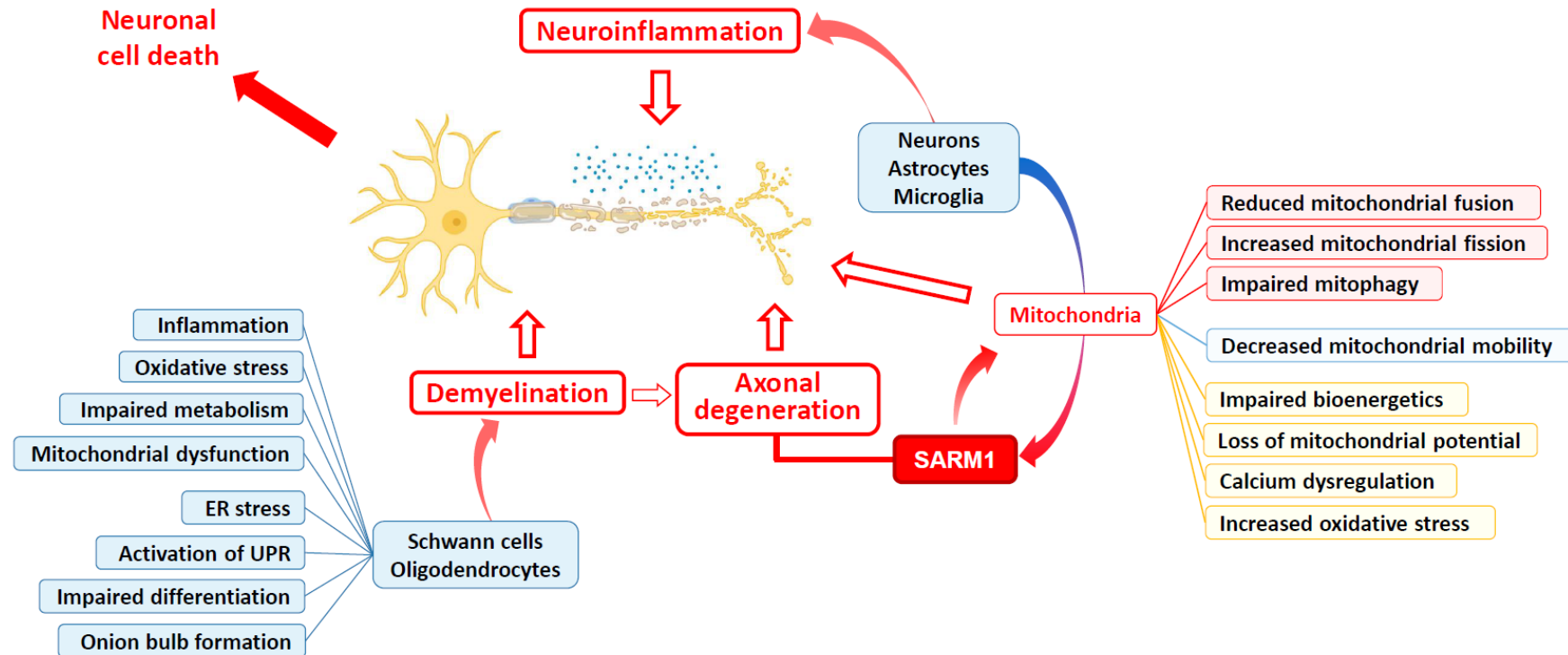
Spinal Motor Neuron in TDP-43* A315T Mice



LMT837 significantly decreases pathological TDP-43 aggregation in spinal motor neurons, demonstrating strong therapeutic potential for ALS therapy.

Broader Application Across Neurodegenerative Disorders

- Therapeutic potential across neurodegenerative diseases by preventing axonal degeneration and demyelination and reducing neuroinflammation within neuronal microenvironments.

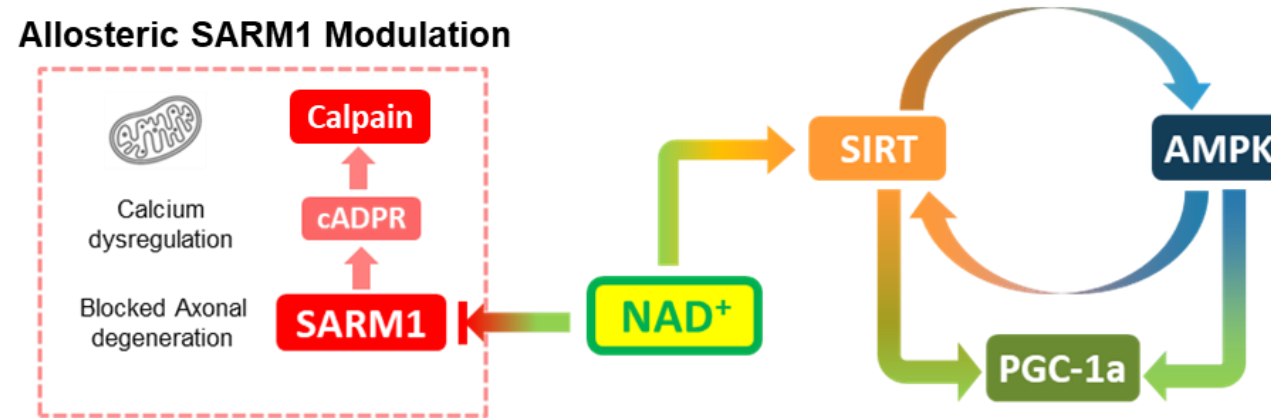


Major Intellectual Properties

Application No.	Title of Invention	Status	Patent No.	Registration Date	Comment
10-2019-0023942	Benzoindazolone Compound and Use Thereof	Granted	10-2201768	6-Jan-2021	LMT503 KR
PCT/KR2020/002459			PCT		US, CA, CN, AU, JP, IN, RU, EP, BR, HK
PCT/KR2021/010842	Imidazoquinoline or Benzoindazolone Compound and Use Thereof	Granted	10-2512398	16-Mar-2023	LMT801, LMT837 KR
			PCT		AU, JP, CN, HK, US, IN, BR, CA, EP
10-2025-0087405	Pyrrole-2,5-dione Compound and Use Thereof	Filed	-	-	LMT3012 KR

- Two composition-of-matter patents have been granted in **2021 (LMT503)** and **2023 (LMT801, LMT837)**, and now are in the process of seeking international protection through the PCT system.
- Application for Composition-of-matter patent of **PKM2 activators (LMT3012)** filed (KR Application No. 10-2025-0087405).

Executive Summary of Lmito's SARM1 Inhibition Program



- Mechanism for Nerve Regeneration
 - Axonal Integrity : Lmito's approach elevates NAD⁺ levels, stabilizing SARM1's inactive state through ARM domain binding and preventing axon degeneration.
 - Re-myelination : Also elevated NAD⁺ improves energy metabolism, DNA repair, mitochondrial function, stress resistance, autophagy, synaptic plasticity, and reduces neuroinflammation.
- LMT837 - ready to initiate GLP-toxicological study
- Actively seeking strategic global partnerships for co-development to advance our innovative pipeline



developing breakthrough drugs for a healthier, happier, and better life

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